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Use of Predictive Road Safety Analytics: Integrating Internal and External Driving and Journey Management Data with Power BI

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Abstract

Driving remains one of the highest-risk activities in oil and gas operations, particularly across expansive geographies with varied infrastructure, security profiles, and driver populations. Traditional reactive safety approaches such as post-incident reviews and stand-alone IVMS reports are insufficient to identify emerging trends and prevent high-potential vehicle incidents. This paper introduces an integrated, data-driven Power BI dashboard developed to analyze and forecast driving risk across nine countries in the MENA and Pakistan regions.

The dashboard unifies internal and external driving data sources including IVMS, AI camera violations, journey management plans (JMP), behavioural safety observations (*Radar*), fatigue tracking, driver utilization hours, and driving real-time user feedback ("Rate My Driving"). The tool visualizes eight critical dimensions of road safety: driving exposure, driving violations, RAG performance, passenger driver evaluations, behavior-based safety, road incidents, fatigue management, and journey management compliance. The dashboard converts raw driving data into actionable safety intelligence, enabling HSE and operations teams to proactively retrain drivers, mitigate high-risk routes, and close compliance gaps.

Piloted across diverse terrains, this solution resulted in measurable improvements: increased journey planning adherence, reductions in over-speed and distraction events, and a cultural shift toward transparent safety ownership. This paper details the architecture, results, and lessons learned from deploying this dashboard and highlights its potential to drive a new standard in predictive road safety across Oilfield operations.

Introduction

Vehicle-related incidents remain among the top contributors to serious injuries and fatalities in the oil and gas sector, particularly in regions characterized by long distance driving, complex logistics, and high environmental risk. Despite advancements in vehicle monitoring systems (IVMS), many companies still struggle with fragmented data, delayed reporting, and the limited ability to intervene before incidents occur.

In high-risk areas such as MENA, and Pakistan, the need for proactive driving safety systems is critical. Long distance, night driving, Hazardous Materials transportation, inadequate road infrastructure, and driver fatigue combine to create an elevated risk environment. These challenges demand a paradigm shift: from

reactive reporting to predictive analytics, where near-real-time data is transformed into forward-looking safety insights.

This paper introduces a unified Power BI dashboard created to address this challenge by visualizing and analyzing the full spectrum of driver behavior, exposure, and performance. The tool provides an operational snapshot and long-term trends across nine categories critical to road safety, enabling field supervisors and HSE professionals to drive timely, data-informed interventions.

Objective

The primary objective of this work is to enhance road safety performance in Oil and Gas operations by designing and deploying a predictive analytics dashboard that integrates key internal and external driving datasets into a single, visual interface. The dashboard is not only intended for monitoring but also for forecasting driver risk by:

- Providing a multi-dimensional overview of driving exposure
- Analyzing journey management plan (JMP) compliance
- Measuring real-time passenger feedback via "Rate My Driving" tools
- Identifying fatigue indicators and consecutive driving hours
- Mapping incident root causes behavioral trends
- Benchmarking drivers through the RAG (Red-Amber-Green) safety classification
- Aggregating behavioral safety cards (*Radar*) for trend detection
- Highlighting deviations, HAZMAT movement, and route risks
- Providing real time alerts triggered by fatigue or distractions from AI Fatigue dash cameras.

By enabling proactive safety management and targeted coaching, the dashboard aims to significantly reduce preventable road incidents and promote a culture of continuous improvement in fleet safety.

Methodology

A data-driven approach was adopted for this initiative. The dashboard was built using Microsoft Power BI, with data sourced from a wide array of systems and sensors:

- IVMS telemetry (speeding, harsh braking, distraction alerts)
- AI camera systems (lane discipline, mobile phone use)
- Journey Management platform (trip risk level, route compliance, driver assignment)
- *Radar* behavior-based safety cards (employee-reported observations)
- "Rate My Driving" survey responses (passenger evaluations after each trip)
- Fatigue monitoring logs (driving hours per day/week/month)
- HSE reporting tools (road incidents, root causes, lessons learned)
- LMS training records (certification status, coaching sessions)

The backend data model included transformation and cleaning via Power Query, with predictive components built in R and Python for scoring high-risk behaviors. These predictive insights were mapped

to individual drivers, vehicle IDs, and regional clusters. Visualization layers were designed to be accessible by field HSE staff, transport supervisors, and senior leadership.

Each of the nine dashboard categories was designed with a specific preventive outcome in mind:

	Category	Focus Area
1	Driving Exposure	Mileage, geography, day/night driving ratio
2	Driving Violations	Speeding, harsh braking, AI-flagged distractions
3	RAG Performance	Driver scoring and country comparisons
4	Rate My Driving	Passenger feedback, improvement trends
5	<i>Radar</i> Program	Safety observations linked to behavior
6	Driving Incidents	Root cause tracking, pattern analysis
7	Fatigue Management	Long shift alerts, driver inactivity, continuous hours
8	Journey Management	JMP compliance, HAZMAT routes, deviation flags
9	AI Cameras	Yawning events, distractions, fatigue

Data was refreshed daily, and the dashboard was deployed across operations in Kuwait, Iraq, Oman, UAE, Tunisia, Qatar, Pakistan, Egypt, Libya and Algeria. This wide scope allowed the team to calibrate the tool against a broad range of terrains, driving conditions, and safety cultures.

Results and Discussion

The dashboard was deployed across ten operational countries: Iraq, Kuwait, Oman, UAE, Tunisia, Qatar, Pakistan, Egypt, Libya and Algeria. These countries present highly varied conditions—ranging from desert highways to high-security zones—each posing unique road safety challenges. Within the first six months of deployment, notable improvements were recorded:

- Driving violations dropped by up to 25% in regions where targeted retraining and coaching were enabled based on dashboard insights.
- Rate My Driving feedback scores improved, with the average rating increasing from 8.5 to 9.07.
- Drivers were identified for fatigue-related risks, leading to route reassignment or enforced rest cycles.
- RAG score improved in all the countries, with several months with zero drivers with Red rating.
- *Radar* cards increased in quantity, signaling increased safety awareness and field-level participation.
- Journey management compliance saw a marked rise, particularly in high-risk trip classifications and HAZMAT transport monitoring.

Perhaps most significantly, the dashboard fostered shared accountability. Supervisors, HSE teams, and drivers began referencing the same visual data to discuss performance, enhancing collaboration and reducing friction in performance management conversations.

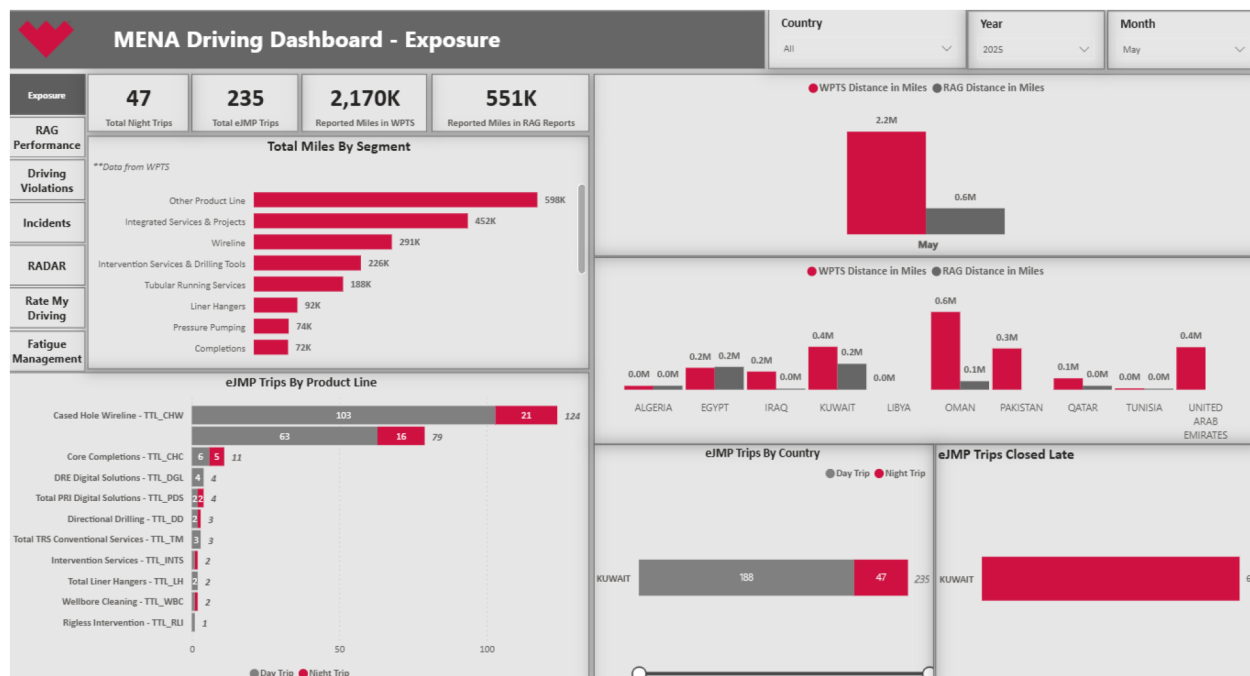
Dashboard Analysis by Category

This section explores the nine dashboard categories in detail, referencing the actual visualizations sample reports. The name of employees was blurred/removed for effects of privacy and only country level analysis is provided, but the dashboard has the capability to drill down analysis per driver in the different categories which is one of the important features of the dashboard.

Driving Exposure

The Driving exposure tab offers a high-level view of driving exposure across countries, including mileage, analysis of trips during the day versus during night, country comparisons. For example:

- The total logged driving hours exceed 69,000.
- Kuwait accounts for 28,830 hours, driven by 165 drivers, making it the highest exposure region.
- The dashboard distinguishes monthly and daily average hours per driver, helping identify over-utilization or underperformance.



The last chart supports the Journey Management Coordinator track the amount of JMPs that are closed late.

By identifying countries with high total hours and prolonged consecutive driving days (e.g., UAE and Kuwait), the dashboard enables more balanced workforce allocation and reduced fatigue risk.

Driving Violations

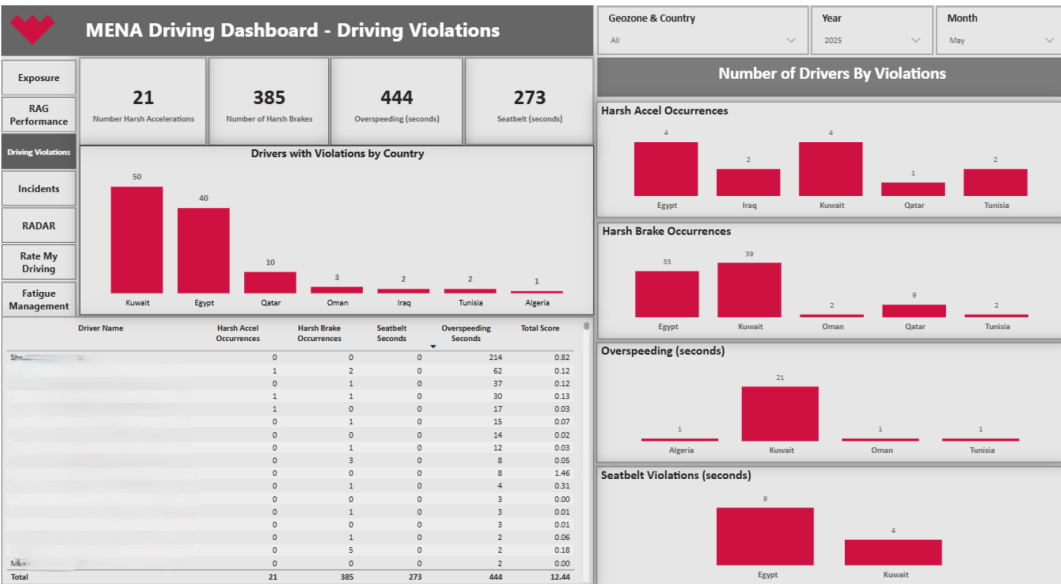
This tab relies on IVMS data such as: speeding, harsh braking, hard cornering, etc.

- Overspeed events by zone
- Harsh deceleration or acceleration patterns
- Seat belt alert.

Trend analysis over time supports driver coaching, while cross-country comparison drives regional accountability.

In some cases, it helped identify upon investigation that the sensors in some seat belts were not properly functioning triggering false alarms, which were rectified and recalibrated by the IVMS vendor.

Insight: Drivers reduced IVMS violations in all countries by focusing on coaching and interventions.

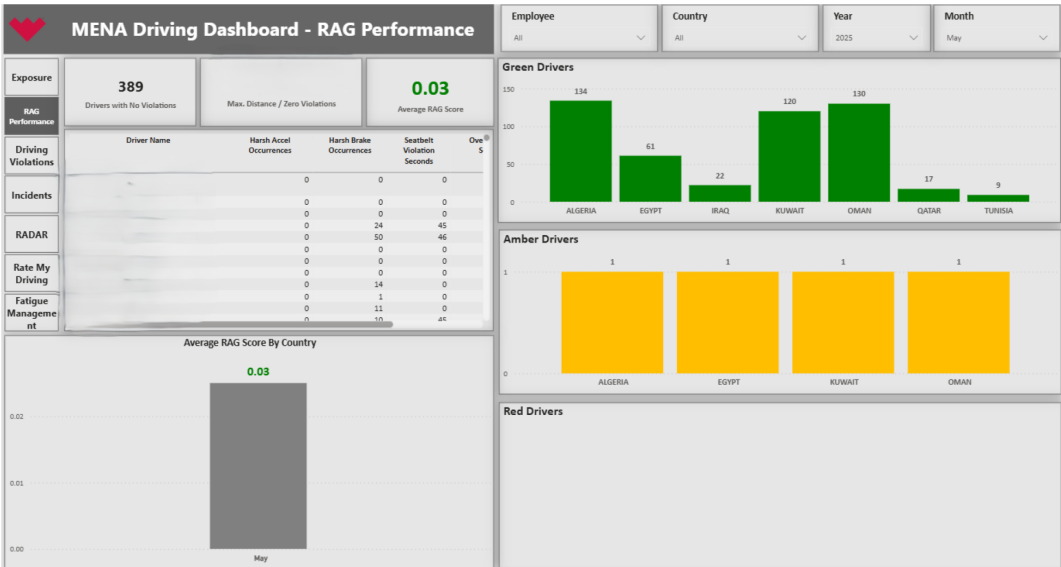


RAG Performance

The RAG Performance tab highlights the best and worst drivers using a clear traffic light system, including driver risk classification (Red/Amber/Green), country scores, avoidance of red-rated drivers. It enables:

- Immediate filtering by country and vehicle
- Benchmarking against the country average score
- Monitoring improvements over time for red-classified drivers

In Tunisia, proactive rotation of high-risk (red) drivers and weekly coaching sessions improved the green driver percentage from 67% to 84% in four months.



Rate My Driving

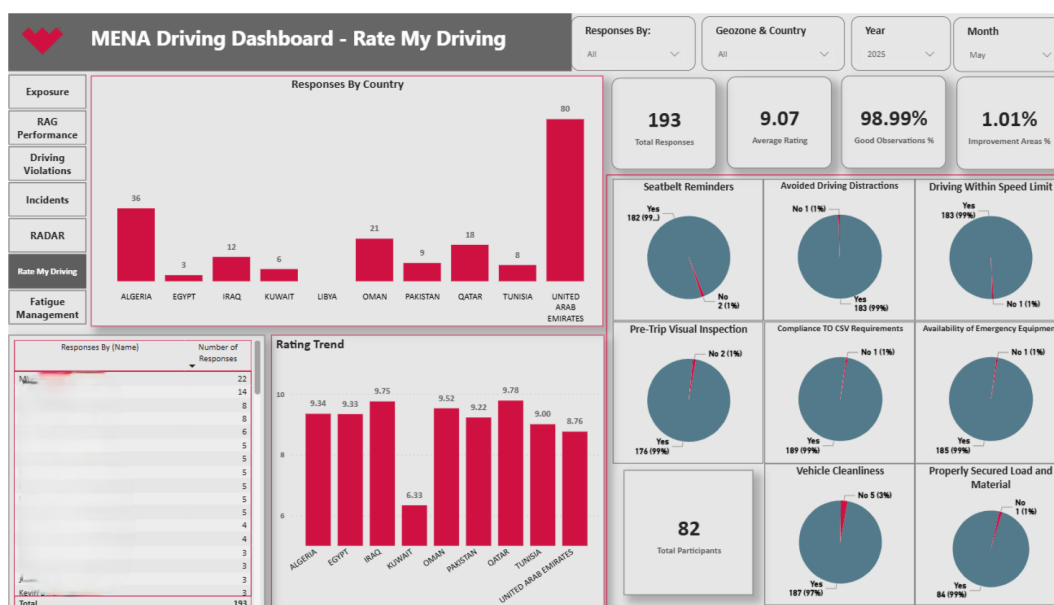
The *Rate My Driving* (RMD) dashboard is a powerful behavior reflection tool. The target is to have every passenger rating the company drivers at the end of each trip by scanning a QR code available in each vehicle

dashboard and to submit immediate feedback that also reaches all company employees, but particularly it is reviewed by the HSE and JM teams. This tab in the dashboard includes passenger evaluations post-trip, Feedback by country, Improvement areas. It aggregates 193 responses, where:

- Average driver rating is 9.07 out of 10
- UAE has the highest number of responses (80), indicating better engagement, with 41.45% of the total responses within MENA.
- Common improvement areas (via pie charts) include: pre-trip inspection, seatbelt reminders, vehicle cleanliness

The feedback loop is immediately submitted by passengers at the end of each journey which increases honesty and operational credibility.

Insight: In Kuwait, low driver ratings (avg. 6.33) triggered retraining of specific individuals. The tool visualized driver-specific ratings and correlated them with violation data to validate coaching needs.



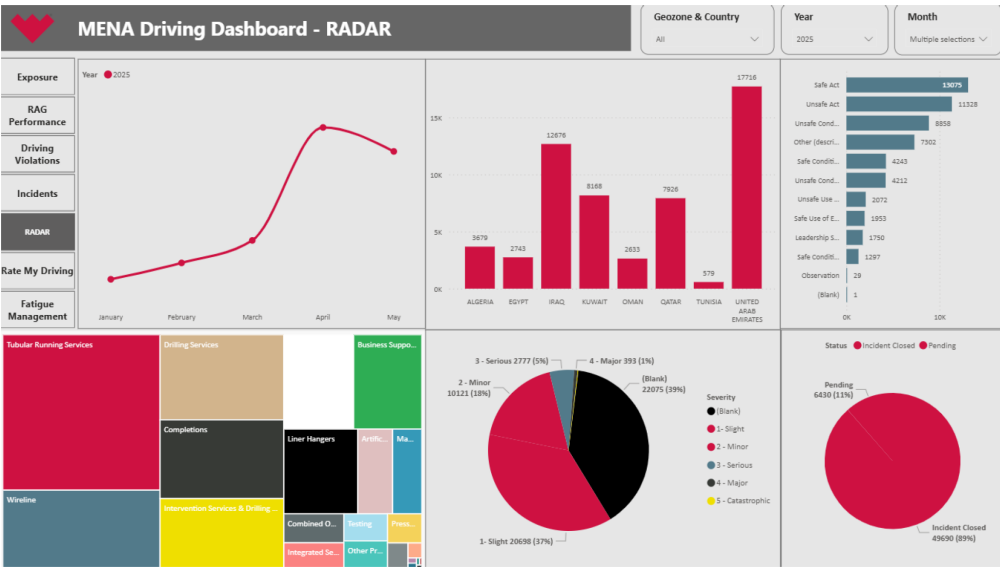
Radar Behavior-Based Program

Radar cards allow any employee to report observed behaviors—positive or negative during driving. This tab includes BBS cards submitted by employees related to driving behavior, near misses, and interventions. Trends are grouped by category:

- Speeding and fatigue near-misses
- Distractions (mobile use, inattentiveness)
- Unsafe overtaking or reversing
- Positive interventions (praise for safe behavior)

Cards are tagged per country and severity, then classified for follow-up by local supervisors.

This section helps analyse the trend on behaviour base safety cards related to driving raised per country, PL, month, severity, etc.



Driving Incidents

The Incident category cross-links historical event data with current dashboard performance. The dashboard tab includes actual road incidents, root causes, if the investigation report is closed. For each incident, analysts review:

- Severity of the driving incidents
- Common root causes
- Countries with more vehicle incidents per severity
- Time taken to investigate a vehicle incident and to close the action plan.

This 360-degree review approach helps identify repeat contributors and refine preventive measures.

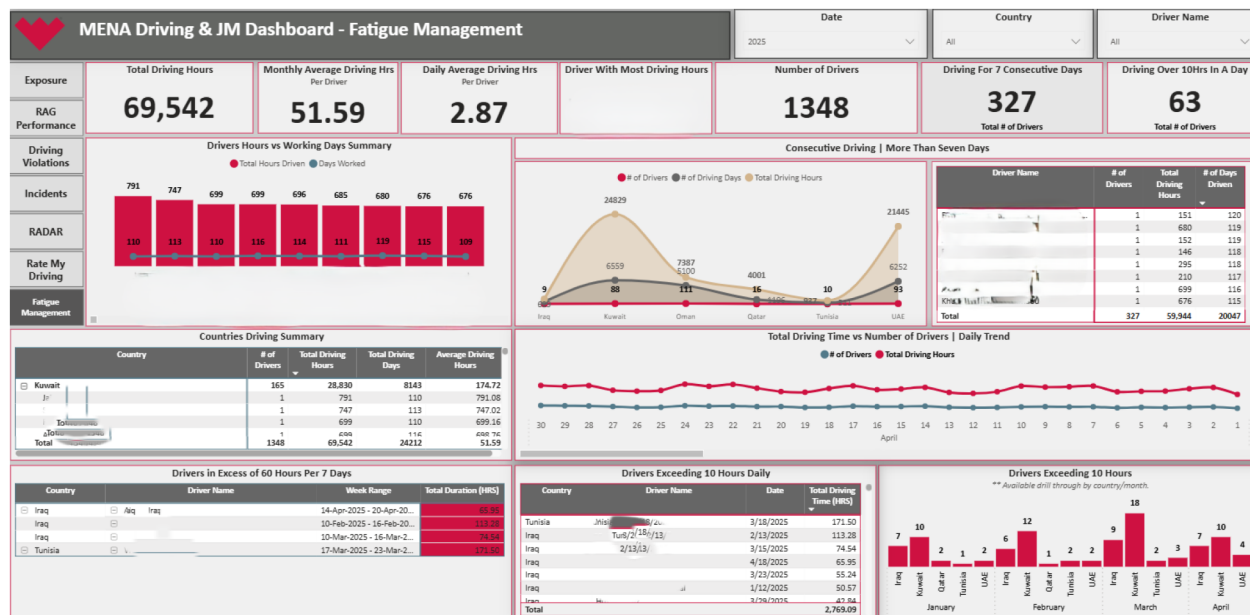


Fatigue Management

This is among the most advanced tabs in the dashboard, and it includes Consecutive days/hours of driving, long daily durations, inactive periods

- Drivers flagged for driving 7 consecutive days
- Drivers exceeding 10 hours in a single day
- Countries with drivers investing more time on the road for consecutive days.

The dashboard allows drill-through by country or driver, enabling fatigue alerts and corrective action.

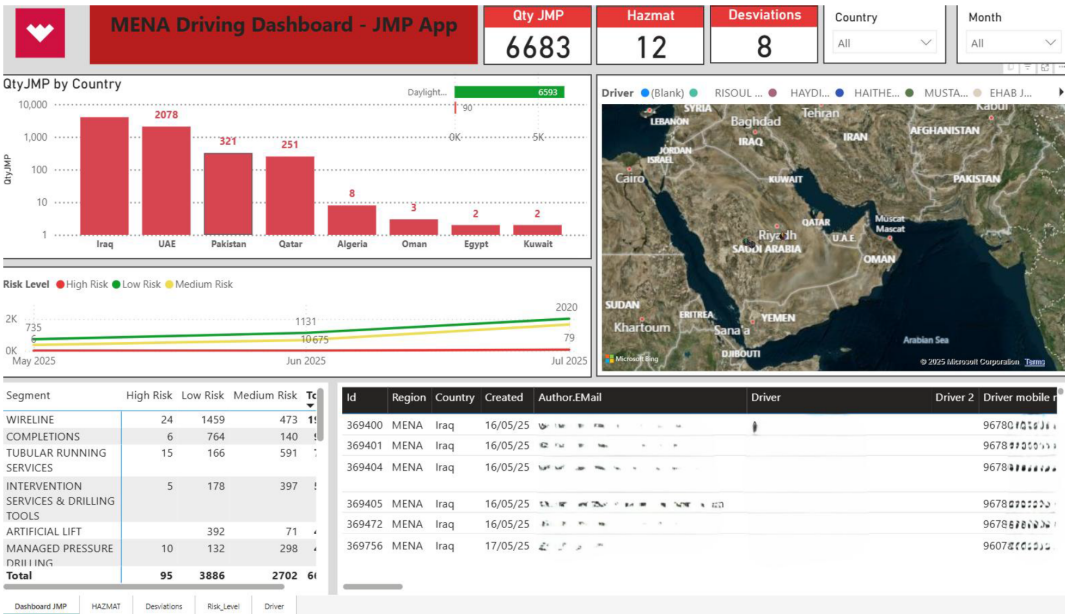


Journey Management

This tab helps analyzing JMP adherence, Trip risk level (low/medium/high), HAZMAT movement, route deviations, and in more detail:

- Comparison between planned vs. actual route
- JMP risk classification adherence (based on HSE matrix)
- Alerts for missing or bypassed approvals
- Concentration of common areas where the vehicle fleet is located in a map
- HAZMAT-specific trips flagged for additional oversight

Supervisors can visualize deviations geographically and evaluate potential non-compliance.

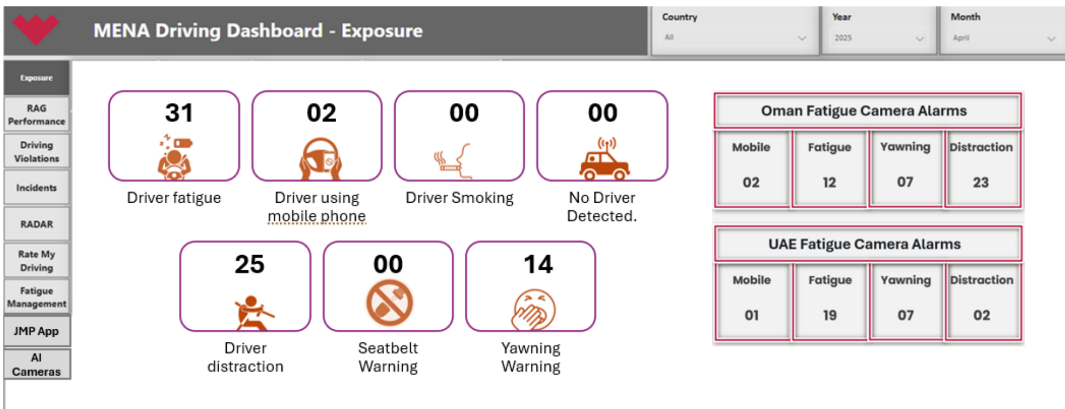


Fatigue AI Dash Cameras

Fatigue AI cameras in vehicles offer significant advantages primarily centered around enhanced road safety and operational efficiency, by detecting driver fatigue and distraction and providing timely alerts to prevent accidents and improve overall driver performance and fleet management. Includes:

- Drivers fatigued (closing eyes)
- Yawning
- Smoking
- Distracted with the mobile phone

This also facilitates incident investigation, as the cameras clearly show the driver’s behavior, so that the investigation does not solely rely on the driver statement.



Innovation

The predictive road safety dashboard represents a major advancement in how driving risk is monitored, understood, and acted upon in the oil and gas sector. Traditionally, driving safety programs rely heavily on post incident reporting, manual journey approvals, and basic IVMS outputs.

All Health, Safety and Environmental professionals need to take advantage of technology to innovate, the key to that innovation is to envision areas that are traditionally analyzed in isolated and independent reports and consolidate all the information into a single dashboard, especially in regions where there are different vendors for driving related solutions, for instance consolidating four different In Vehicle Monitoring Systems (IVMS) vendors from the 10 countries within MENA and make a unified visual report where countries can be compared with the same settings to analyze driving performance. The same approach is applied to all other aspects generating data to analyze proactive and reactive driving performance.

This dashboard, however, offers several innovative breakthroughs:

- Multi-source data integration: By combining IVMS, AI camera data, JMP systems, "Rate My Driving" passenger surveys, fatigue indicators, and behavioral cards (Radar), the dashboard creates a single source of information. This holistic view allows safety professionals to triangulate risk factors.
- Predictive fatigue modeling: By analyzing hours driven across days, weeks, and months—alongside inactivity periods—the system flags at-risk drivers before they reach fatigue thresholds.
- Real-time feedback loop: The "Rate My Driving" tool empowers passengers to immediately evaluate their drivers, democratizing safety accountability. This input is visible on the dashboard within hours of the trip, enabling rapid corrective action.
- Visual coaching aids: Supervisors can download or screenshot specific visual insights (such as a driver's 7-day fatigue history or violation map) and use them in direct conversations with drivers during training or incident investigations.
- Localization with global oversight: Although country managers view their specific performance, the platform also enables corporate-level benchmarking and trend analysis across operations.

This innovation shifts the company's safety posture from reactive to predictive and preventive, reducing both lagging and leading indicators.

Challenges and Limitations

Despite its effectiveness, the implementation of the dashboard faced several challenges:

- Data accuracy and completeness: IVMS coverage varied between countries, and not all vehicles had uniform device calibration. To accomplish uniformity it is required that all IVMS vendors are willing to use the same settings and make constant recalibrations to ensure there is uniformity on the data; for instance ensuring that 0.03 RAG report in a country is the same than in a different country using a different IVMS vendor with a monthly RAG score of 0.03.
- The Fatigue Artificial Intelligence (AI) cameras have only been installed in 2 countries showing an early analysis but the expectation is that if more countries are allowed to install this AI cameras the system is ready as "plug and play" to increase data points and opportunities to compare trends, performance, challenges, etc.
- Different IVMS providers with different systems, IT security, and willingness to provide the API key to automatically obtain the data for analysis. This process might take several weeks or months of seeking approvals internally within the company to build a dashboard with different data that the existing internal systems or applications, as well as convincing vendors to provide accessibility to the API key to automatically get access to data.
- Client or regulatory restrictions. The Fatigue AI cameras are not allowed to be installed in all countries within the Middle East and North Africa (MENA) and Pakistan, so the analysis in this paper is limited to a few countries only, currently available in Oman and UAE. High possibilities to get government approval in Egypt, and Pakistan.

- Data silos and integration delays: Bringing together systems from different functions such as Fleet Management or Transport; Health, Safety and Environment (HSE), Human Resources (HR), Information Technology (IT) and Operations required coordinated data governance and periodic reconciliation to resolve mismatches.
- Technical skills gap: While Power BI is user-friendly, some field supervisors lacked data literacy. Targeted training sessions were necessary to improve dashboard interpretation. The more complex a dashboard become with multiple sources of data, it might become a big concern if some sections of the dashboard stop automatically synchronizing or need to be troubleshoot and the person or owner of the dashboard suddenly leaves the organization.
- Regional cultural differences: Feedback and *Radar* card submission rates varied significantly due to local norms, requiring country-specific engagement strategies.

Addressing these limitations involved continuous improvement of the dashboard backend, localized rollout plans, and strong alignment between safety and operations leadership.

Conclusion and Recommendations

This paper demonstrated how a unified Power BI dashboard that integrates predictive analytics and multiple data sources can significantly improve road safety in the Oil and Gas industry. By organizing data into eight strategic categories ranging from exposure and fatigue to journey management and behavior-based feedback, the platform empowers supervisors and drivers alike to take timely, informed actions.

The initiative has already led to measurable safety improvements: fewer high-risk drivers, less report with a red score in the monthly RAG performance report, better journey plan adherence with less JMP deviations, reduction in night driving trips due to better planning, better identification of best drivers in different countries based on a more rounded criteria and not limited only to RAG reports – this is very relevant for award and recognition program, reduction on the occurrence of driving related incidents and for those vehicle incidents that happen to be reduced in severity, and finally an enhanced cultural ownership of safety performance. The dashboard facilitated a shift in mindset from lagging to leading indicators.

Based on these outcomes, the following recommendations are proposed:

1. Scale horizontally: Expand the dashboard to more aspects related to driving, for instance: Defensive Driving training validity.
2. Integrate mobile access: Allow supervisors and drivers to view and interact with their scores and feedback on mobile devices.
3. Automate alerts: Build in threshold-based notifications (e.g., driving >10 hours/day triggers fatigue alert) that notify supervisors via email or Teams.
4. Close the loop with coaching logs: Track not just driver violations, but also whether corrective actions (e.g., training, warnings) have been completed.
5. Leverage AI for journey risk: Use AI models to classify trip risk dynamically based on historical incident data, rather than only relying on pre-defined JMP classifications.

This dashboard represents a foundational step toward a data-informed, people-centered safety culture where technology augments human judgment and strengthens preventive action.

In a sector where transportation incidents remain among the highest contributors to serious injuries and fatalities, this work presents a scalable and cost-effective method to modernize fleet safety oversight. As oil and gas companies continue to adopt digital transformation, embedding tools like this into their operational fabric will be key to sustainable HSE performance improvement.

Future work will focus on incorporating machine learning to predict at-risk trips before departure, integrating biometric fatigue indicators, and extending dashboard access to subcontractor fleets for broader impact.

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